

SOFTWARE DESIGN & ANALYSIS DOCUMENT (SDAD)

BarangayHealth — Community Health Monitoring System

Aligned with UN SDG 3: Good Health and Well-Being

Course: PC11 — System Analysis and Design | BSIT 2nd Year | 2nd Semester | AY 2025–2026

Field	Detail
Document Type	Software Design and Analysis Document (SDAD)
Project Title	BarangayHealth — Community Health Monitoring System
SDG Alignment	UN SDG Goal 3: Good Health and Well-Being
Author / Student	Solo Submission — BSIT 2nd Year
Course Code	PC11 — System Analysis and Design
Version	1.0 Final
Date Submitted	May 16, 2026
Companion Document	01_SRS_Final.pdf

1. PROCESS ARCHITECTURE — DATA FLOW DIAGRAMS (DFD)

1.1 Context Diagram — Level -1 DFD (Gane-Sarson Notation)

The Context Diagram represents the entire BarangayHealth system as a single process bubble surrounded by all external entities and data flows. This is the highest level of abstraction using Gane-Sarson notation.

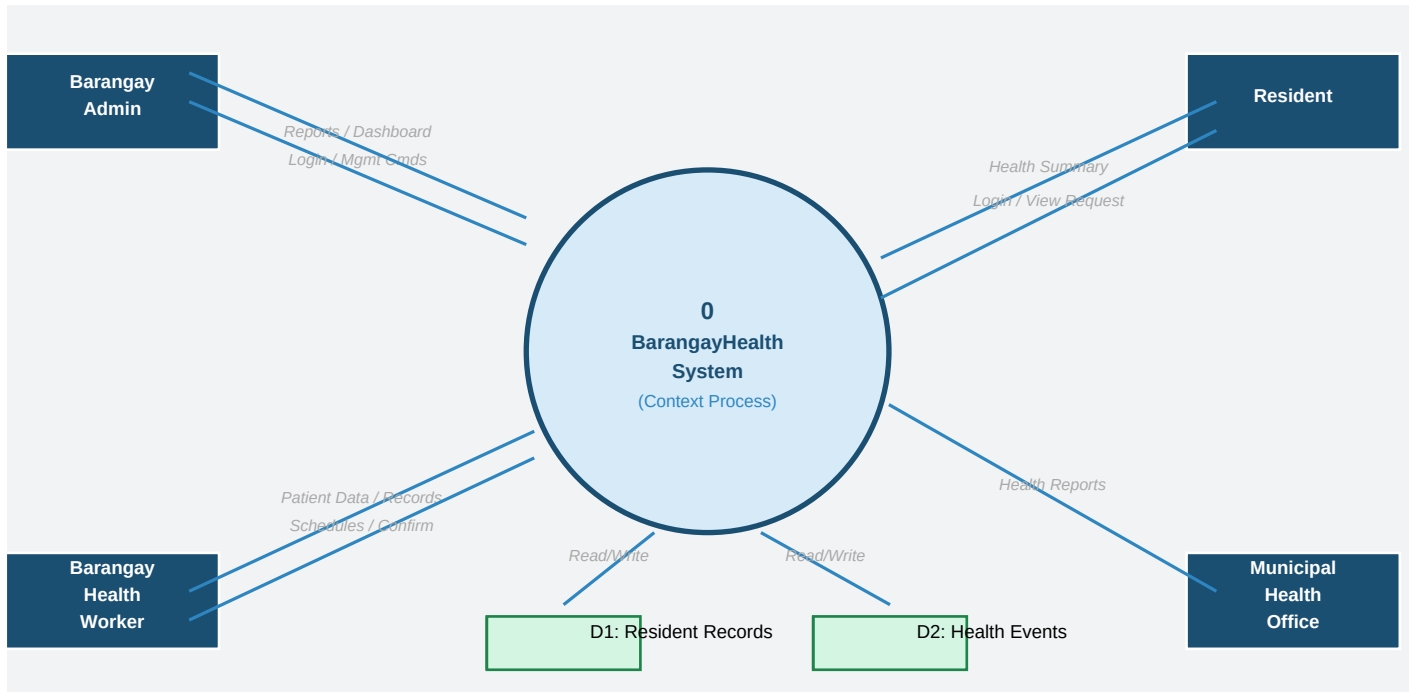


Figure 1. BarangayHealth Level-0 / Context DFD (Gane-Sarson Notation)

1.2 Level-0 DFD — Major Process Decomposition

At Level-0, the central process is decomposed into the following major subsystem processes. Each process communicates with shared data stores.

Proc ess #	Process Name	Input Data Flows	Output Data Flows	Data Stores Used
P1	User Authentication & RBAC	Login credentials	Authenticated session / Access denied	D0: USER
P2	Resident Profile Management	New/updated resident data (BHW)	Resident profile confirmation	D1: RESIDENT
P3	Immunization Tracking	Vaccine details, date (BHW)	Immunization record, next due date	D1, D2: IMMUNIZATION
P4	Prenatal Care Monitoring	Prenatal visit data (BHW)	Visit record, follow-up schedule	D1, D3: PRENATAL_VISIT
P5	Consultation Management	Consultation notes (BHW)	Consultation record, follow-up date	D1, D4: CONSULTATION

Proc ess #	Process Name	Input Data Flows	Output Data Flows	Data Stores Used
P6	Report Generation Engine	Report parameters (Admin)	PDF/CSV Monthly Health Summary	D2, D3, D4, D5: REPORT
P7	Notification Service	Trigger from P3/P4 (internal)	Appointment reminder → Resident	D1: Resident contact info

2. DATA ARCHITECTURE — ENTITY-RELATIONSHIP DIAGRAM (ERD)

2.1 Logical ERD — Crow's Foot Notation (Normalized to 3NF)

The Logical ERD below represents the relational data model for BarangayHealth. All entities are normalized to Third Normal Form (3NF): each non-key attribute depends solely on the primary key. Crow's Foot notation is used throughout.

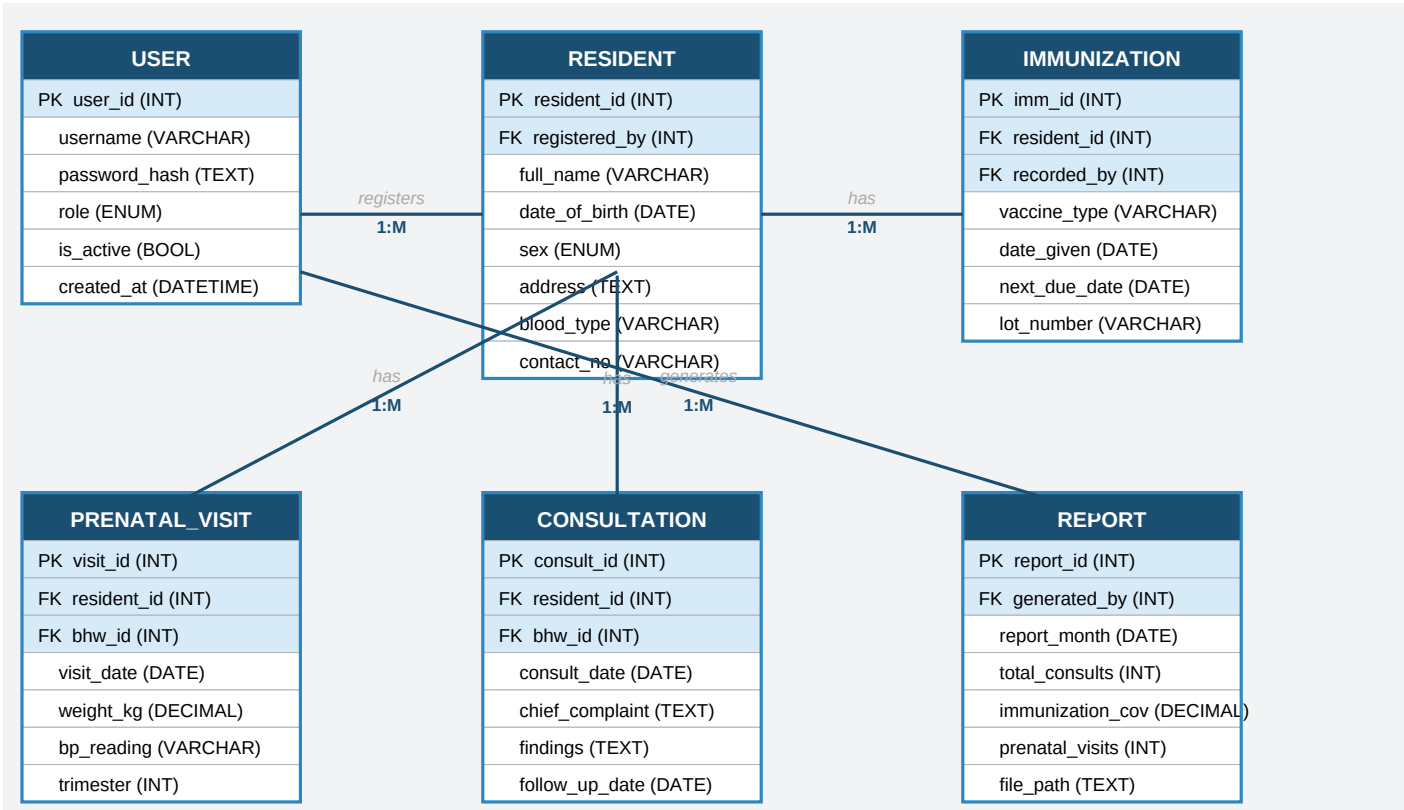


Figure 2. BarangayHealth Logical ERD (Crow's Foot Notation, 3NF)

2.2 Entity Descriptions and Cardinality

Entity	Primary Key	Description	Relationship
USER	user_id	System users: Admin and BHW roles	1 USER registers M RESIDENTs; generates M REPORTs
RESIDENT	resident_id	Community member health profile	1 RESIDENT has M IMMUNIZATIONs, PRENATALs, CONSULTs
IMMUNIZATION	imm_id	Individual vaccine event per resident	M:1 to RESIDENT; references USER (BHW)
PRENATAL_VISIT	visit_id	Maternal health visit record per resident	M:1 to RESIDENT; references USER (BHW)
CONSULTATION	consult_id	General consultation record per resident	M:1 to RESIDENT; references USER (BHW)
REPORT	report_id	Monthly aggregate health report per barangay	M:1 to USER (Admin who generated it)

3. SYSTEM INFRASTRUCTURE — 3-TIER ARCHITECTURE

3.1 3-Tier Architecture Overview

BarangayHealth follows a classic 3-Tier Architecture to enforce separation of concerns, improve maintainability, and allow independent scaling of each layer.

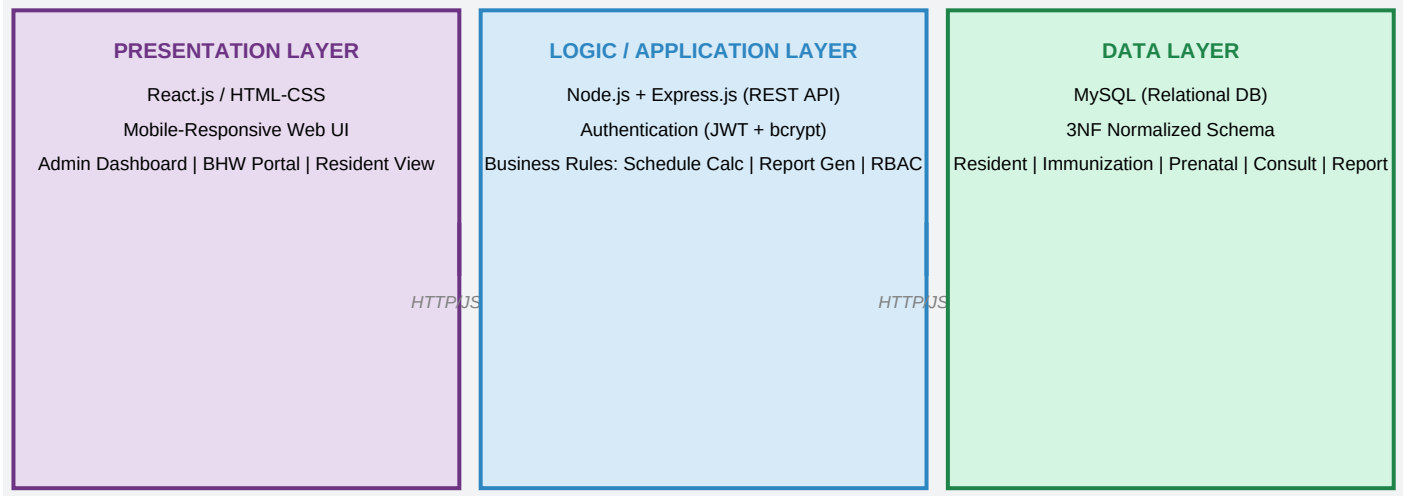


Figure 3. BarangayHealth 3-Tier Architecture Sketch

Tier	Technology Stack	Responsibility
Presentation Layer	React.js + Tailwind CSS (Web)	UI rendering, user interaction, form validation, mobile-responsive views for Admin, BHW, Resident portals.
Application Layer	Node.js + Express.js REST API	Business logic: RBAC enforcement, next-immunization-date calculation, report aggregation, JWT authentication, bcrypt password hashing.
Data Layer	MySQL 8.0 + Sequelize ORM	Persistent storage of all health records. Schema enforces 3NF, referential integrity (FK constraints), and indexed queries for fast retrieval.

4. TELOS FEASIBILITY ANALYSIS

Feasibility	Assessment	Justification
Technical	FEASIBLE	All technologies (React, Node.js, MySQL) are open-source, free, and well-documented. The system requires only a standard web server and MySQL DB — no specialized hardware. BSIT-level competency is sufficient for implementation.
Economic	FEASIBLE	Total estimated deployment cost: PHP 0 (development) + PHP ~5,000/year (cloud hosting via DigitalOcean Droplet or local server). No licensed software required. Operational savings from eliminated paper-based records justify the cost.
Legal	FEASIBLE (with Compliance)	System design incorporates Data Privacy Act (RA 10173) compliance: RBAC restricts PHI access, audit logs are maintained, and no resident data is shared externally without written consent. HIPAA-equivalent practices adopted locally.
Operational	FEASIBLE	BHWs are trainable within 1–2 sessions. A printed Quick Reference Guide will be provided. System UI is designed for low-literacy users with icon-based navigation. Offline mode with sync reduces dependency on stable internet.
Scheduling	FEASIBLE	Analysis & Design phase completed within 2nd semester (Jan–May 2026). A working prototype can be developed in 8–10 weeks post-approval. Solo scope is limited to core modules (P1–P6) per MoSCoW prioritization.

5. UI/UX BLUEPRINT — CORE USER JOURNEY WIREFRAMES

5.1 Wireframe Descriptions (High-Fidelity Mockups)

The following wireframes define the core user journeys for each role. Final high-fidelity mockups are located in the PROTOTYPE/ folder of the GitHub repository.

Screen ID	Screen Name	Role	Key UI Elements	Core Action
WF-01	Login Page	All	Username/password fields, role indicator, 'Sign In' button, forgot password link	Authenticate user
WF-02	Admin Dashboard	Admin	Summary cards: Active Residents, Pending Immunizations, Monthly Consults; Quick links; Alert panel	View system overview
WF-03	Resident Registration	BHW	Multi-step form: Personal Info → Medical History → Contact; Progress bar; Validation highlights	Add new resident
WF-04	Resident Profile View	BHW/Admin	Profile header, tabbed sections: Immunizations Prenatal Consultations; 'Add Record' buttons per tab	View/update health records
WF-05	Immunization Entry Form	BHW	Dropdown: vaccine type; Date picker; Lot number field; 'Next Due' auto-display after submit	Log immunization event
WF-06	Report Generation Page	Admin	Month/year selector; 'Generate Report' button; Preview table; Download PDF/CSV buttons	Generate monthly report
WF-07	Resident Health Portal	Resident	Read-only: Personal summary; Upcoming appointments; Immunization history; 'Contact BHW' button	View own health record

6. VALIDATION PLAN — REQUIREMENTS TRACEABILITY MATRIX (RTM)

6.1 Requirements Traceability Matrix

The RTM links each Functional Requirement (from the SRS) to its corresponding Use Case, DFD Process, ERD Entity, Wireframe, and Test Case. Zero 'dead-end' requirements — every FR is traceable end-to-end.

FR-ID	Requirement Summary	Use Case	DFD Process	ERD Entity	Wireframe	Test Case ID
FR-01	Admin manages user accounts	UC-Admin	P1	USER	WF-02	TC-01
FR-02	BHW registers resident health profiles	UC-01	P2	RESIDENT	WF-03	TC-02
FR-03	BHW records immunization events	UC-02	P3	IMMUNIZATION	WF-05	TC-03
FR-04	System auto-computes next immunization	UC-02	P3	IMMUNIZATION	WF-05	TC-04
FR-05	BHW records prenatal consultation visits	UC-Pren	P4	PRENATAL_VISIT	WF-04	TC-05
FR-06	Resident views own health record	UC-Res	P1, P2	RESIDENT	WF-07	TC-06
FR-07	Admin generates Monthly Health Report	UC-03	P6	REPORT	WF-06	TC-07
FR-08	In-app appointment notifications	UC-Notif	P7	RESIDENT	WF-07	TC-08
FR-09	Admin/BHW dashboard with live counts	UC-Dash	P6	All aggregates	WF-02	TC-09
FR-10	Export reports to CSV	UC-03	P6	REPORT	WF-06	TC-10

6.2 Sample Test Cases

TC-ID	Test Case Description	Input	Expected Output	FR Ref
TC-02	Register valid resident profile	All required fields filled	Resident saved; unique ID assigned	FR-02
TC-03	Record BCG immunization for infant resident	Vaccine=BCG, Date=valid	Event saved; next due = +4 weeks	FR-03
TC-04	Auto-compute next due date for Hepatitis B	Vaccine=HepB, 1st dose date	Next dose auto-calculated and displayed	FR-04
TC-07	Generate Monthly Report for April 2026	Month=Apr, Year=2026	PDF downloaded with correct aggregate stats	FR-07
TC-06	Resident logs in and views immunization history	Valid resident credentials	Health summary displayed (read-only)	FR-06

